

Manuel Magana

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EDUCATION

Northeastern University

M.S. Computer Science

San Jose, CA

Graduation: May 2026

Relevant Courses: Foundations of Software Engineering, Machine Learning, Deep Learning, Algorithms, Building Scalable Distributed Systems, Computer Networking, Program Design Paradigms

University of California, Berkeley

B.A. Molecular and Cell Biology: Neurobiology

Berkeley, CA

May 2021

Relevant Courses: Data Structures, Machine Structure, Principles & Techniques of Data Science

TECHNICAL SKILLS

Languages: Python, Go, Java, SQL, Bash

Frameworks, Libraries, and Tools: FastAPI, SQLAlchemy, Pydantic, pytest, PySpark, Docker, Redis, DuckDB, AWS, Claude API

EXPERIENCE

Genentech

Bioinformatics Scientist

San Francisco, CA

Nov 2022 – Jul 2025

- Designed and shipped a Python library for automated data validation, integrated into 5 production pipelines and reducing manual review from days to hours.
- Architected a coordinator-worker batch system with task queuing and concurrent execution, enabling parallel processing of large-scale analysis jobs.
- Optimized PySpark ETL pipelines processing 200GB+ mass spectrometry files using partition tuning, broadcast joins, and Parquet caching, reducing batch runtime from hours to minutes.
- Containerized 5+ data pipelines with Docker and deployed them to an internal orchestration platform, enabling reproducible, one-command execution across environments.
- Performed comparative analyses across 4+ proteomics platforms (OLINK, ELLA, ELISA, Mass Spectrometry), streamlining method selection and improving QC processes for multiple projects.
- Developed Python, Bash, and R workflows to automate ad hoc and reproducible analysis of experimental datasets.

Endless West

Junior Data Scientist

San Francisco, CA

Mar 2021 – Nov 2021

- Built ML training pipelines for sensory attribute prediction from chromatography data, with model outputs used to prioritize R&D synthesis targets.
- Developed web scraper in Python using BeautifulSoup to extract and structure chemical property data from public databases enabling quicker reference for R&D teams.

PROJECTS

WhatToDo | AI Event Discovery Platform | *Python, FastAPI, Claude API, Docker, AWS* | [GitHub](#) | [Live](#)

Fall 2025

- Built a dual-provider AI recommendation engine (Claude Sonnet 4 + GPT-4o) with an abstract base class for provider-agnostic event discovery and live web search.
- Implemented JWT auth with bcrypt hashing and a 3-model SQLAlchemy schema backed by AWS RDS (PostgreSQL) in production.
- Deployed containerized API to AWS (App Runner, ECR) with a GitHub Actions CI/CD pipeline.
- Led all backend and infrastructure development for a 4-person team and achieved 96% pytest coverage across auth, AI services, and the event planner.

Physiological Signal Analytics Platform | *Python* | [API](#) | [Library](#)

Sept 2025 – Jan 2026

- Architected a two-layer system: a Python ETL/analysis library for batch processing and a production REST API for external and visualization-driven access.
- Built a FastAPI service over DuckDB exposing raw signals and derived features with offset pagination, composable query filters, and Pydantic validation.
- Implemented analysis endpoints for PCA, FFT, and spectral density supporting interactive exploration of physiological signals.

Rate Limiter Service | *Go, Redis, Docker* | [GitHub](#)

Jan 2026

- Built a high-performance Go reverse proxy implementing fixed window and token bucket rate limiting with configurable per-user quotas.
- Implemented reliable atomic token consumption using Redis transactions to prevent race conditions, ensuring scalable performance under 1,000+ concurrent requests.
- Built a CSV-driven load test harness validating throttle accuracy across burst, steady-state, and multi-user traffic patterns.

Paxos Key-Value Store | *Java, RPC* | [GitHub](#)

Apr 2025

- Implemented a fault-tolerant distributed key-value store using Paxos consensus with full leader election and log replication.
- Designed an RPC-based node discovery and membership protocol enabling dynamic cluster joins with consistent state propagation.
- Validated system reliability and consistency under chaos testing: 10% acceptor failure rate, leader crashes, and network partitions.

PUBLICATIONS

MSstats+ enhances differential analysis in proteomic experiments | *bioRxiv* 2025

- Co-author. [doi:10.1101/2025.09.11.675573](https://doi.org/10.1101/2025.09.11.675573)